Learning Objectives 8.1, 8.2: I can articulate the significance of the displacement current and calculate it, given a time-varying electric field. I can articulate the meaning of and calculate the relaxation time for a variety of materials.

Show your work. Write a **Documentation** line at the end (**None** if you did not collaborate).

1. **A coil in a changing field.** A flat square coil of $N = 50$ turns, 10 cm on a side, lies in the xy plane in a uniform field $\mathbf{B} = \hat{\mathbf{z}}\, B_0 \cos\omega t$ with $B_0 = 50$ mT at 60 Hz. We traverse the coil counterclockwise as seen from $+z$.

   (a) Find the flux through one turn and the amplitude of the coil's EMF.

   The normal is $\hat{\mathbf{z}}$ and the area is $A = 0.1^2 = 0.01$ m^2:

   $$\Phi = B_0 A \cos\omega t = 0.5 \cos\omega t \ \text{mWb}$$

   $$\oint \mathbf{E}\cdot d\mathbf{l} = -N\frac{d\Phi}{dt} = N\omega B_0 A \sin\omega t$$

   $$N\omega B_0 A = 50 \times 377 \times 0.05 \times 0.01 = 9.42 \text{ V}$$

   Φ amplitude = 0.5 mWb

   EMF amplitude = 9.42 V

   (b) At an instant when $\mathbf{B}$ points along $+\hat{\mathbf{z}}$ and is growing, which way does the induced current circulate, seen from $+z$? Explain with Lenz's law and with the sign of the EMF.

   The flux along $+\hat{\mathbf{z}}$ grows, so Lenz's law requires an induced field along $-\hat{\mathbf{z}}$ inside the coil, which a clockwise current produces. The EMF is $-N\, d\Phi/dt < 0$ around the counterclockwise path, so the current runs opposite to the path: clockwise. The two answers agree.

   (c) Starting from the integral form of Faraday's law for a path that does not move, derive $\nabla \times \mathbf{E} = -\partial\mathbf{B}/\partial t$.

   Stokes' theorem turns the circulation into a surface integral, and for a fixed surface the time derivative acts only on $\mathbf{B}$:

   $$\int_S (\nabla \times \mathbf{E}) \cdot d\mathbf{s} = -\frac{d}{dt}\int_S \mathbf{B}\cdot d\mathbf{s} = -\int_S \frac{\partial \mathbf{B}}{\partial t}\cdot d\mathbf{s}$$

   This holds for every surface, however small, so the integrands are equal: $\nabla \times \mathbf{E} = -\partial\mathbf{B}/\partial t$.

2. **A moving bar and a resistor.** A bar of length $\ell = 0.3$ m slides at $v = 4$ m/s along two rails in a uniform field $B = 0.5$ T into the page. A $2\,\Omega$ resistor closes the rails, and the bar and rails have no resistance.

   (a) Find the EMF and the current.

   $$\text{EMF} = vB\ell = 4 \times 0.5 \times 0.3 = 0.6\ \text{V}$$
   $$I = \frac{0.6}{2} = 0.3\ \text{A}$$

   EMF = 0.6 V

   I = 0.3 A

   (b) The magnetic force $I\boldsymbol{\ell} \times \mathbf{B}$ of Lesson 28 acts on the current in the bar. Find its magnitude and say which way it points.

   F = 0.045 N

   $$F = I\ell B = 0.3 \times 0.3 \times 0.5 = 0.045\ \text{N}$$

   The force points against the motion, as Lenz's law requires: the induced current resists the change that produces it.

   (c) Show that the mechanical power needed to keep the bar moving equals the power the resistor dissipates.

   $$Fv = 0.045 \times 4 = 0.18\ \text{W}, \qquad I^2R = 0.3^2 \times 2 = 0.18\ \text{W}$$

   The work done pushing the bar becomes heat in the resistor.

3. **Displacement current in a capacitor.** A parallel-plate capacitor has plates of area $A = 2$ cm^2 separated by $d = 0.5$ mm of a dielectric with $\epsilon_r = 4$. A 100 MHz source applies $v = 5\cos\omega t$ V. Ignore fringing.

   (a) Find the amplitude of the displacement current density between the plates.

   J_d amplitude = 223 A/m^2

   $E = v/d$ has amplitude $E_0 = 5/(5\times10^{-4}) = 10^4$ V/m:

   $$J_d = \epsilon\frac{\partial E}{\partial t} = -\omega\epsilon E_0 \sin\omega t$$
   $$\omega\epsilon E_0 = 2\pi\times10^8\times4\times8.854\times10^{-12}\times10^4$$
   $$= 223 \text{ A/m}^2$$

   (b) Find the total displacement current between the plates, and compare it with the current $C\,dv/dt$ in the wire.

   i_d amplitude = 44.5 mA

   $$i_d = J_d A = 223\times2\times10^{-4} = 44.5 \text{ mA}$$
   $$C = \frac{\epsilon A}{d} = 14.2 \text{ pF}, \qquad \omega C V_0 = 44.5 \text{ mA}$$

   The two are equal.

   (c) Explain, using two surfaces that share one path around the wire, why Ampere's law needs the displacement current to give a single answer here.

   A flat surface that the wire crosses carries the conduction current i. A surface with the same edge that passes between the plates carries no conduction current. Without the displacement current, the circulation of **H** around the path would have two values. With it, the second surface carries $d/dt\int\mathbf{D}\cdot d\mathbf{s} = dQ/dt = i$, and both surfaces give i.

4. **Conduction or displacement current?** Wet soil has $\sigma = 10^{-2}$ S/m and $\epsilon_r = 10$.

(a) For a field $E_0 \cos \omega t$, derive the ratio of the amplitudes of the conduction and displacement current densities.

$$J_c = \sigma E_0 \cos \omega t, \qquad J_d = \epsilon \frac{\partial E}{\partial t} = -\omega \epsilon E_0 \sin \omega t, \qquad \frac{\sigma E_0}{\omega \epsilon E_0} = \frac{\sigma}{\omega \epsilon}$$

(b) Find the ratio at 1 MHz and at 1 GHz, and the frequency at which the two currents are equal.

$$\left.\frac{\sigma}{\omega\epsilon}\right|_{1\text{ MHz}} = \frac{10^{-2}}{2\pi \times 10^6 \times 10 \times 8.854 \times 10^{-12}} = 18.0$$

$$\left.\frac{\sigma}{\omega\epsilon}\right|_{1\text{ GHz}} = 0.0180$$

$$f = \frac{\sigma}{2\pi\epsilon} = 18.0 \text{ MHz}$$

1 MHz:	18.0
1 GHz:	0.0180
equal at	18.0 MHz

(c) Is the soil a good conductor or a good insulator at each frequency?

At 1 MHz conduction current dominates by a factor of 18, so the soil behaves as a conductor. At 1 GHz displacement current dominates by a factor of 56, so the soil behaves as a lossy dielectric.

5. **Relaxation time.** A uniform material has $\sigma = 10^{-4}$ S/m and $\epsilon_r = 3$. At $t = 0$ a free charge density ρ_0 sits inside it.

   (a) Starting from the continuity equation, derive $\rho_v(t)$.

   With $\mathbf{J} = \sigma\mathbf{E}$, $\mathbf{D} = \epsilon\mathbf{E}$, and Gauss's law $\nabla\cdot\mathbf{D} = \rho_v$:

   $$-\frac{\partial \rho_v}{\partial t} = \nabla\cdot\mathbf{J} = \frac{\sigma}{\epsilon}\nabla\cdot\mathbf{D} = \frac{\sigma}{\epsilon}\rho_v \quad \Longrightarrow \quad \rho_v = \rho_0\, e^{-t/\tau_r}, \qquad \tau_r = \frac{\epsilon}{\sigma}$$

   (b) Find the relaxation time τ_r, and the time for the charge inside to fall to 1% of ρ_0.

   $$\tau_r = \frac{3 \times 8.854 \times 10^{-12}}{10^{-4}} = 0.266\ \mu\text{s}$$
   $$t_{1\%} = \tau_r \ln 100 = 4.61\tau_r = 1.22\ \mu\text{s}$$

   τ_r = $0.266\ \mu$s

   $t_{1\%}$ = $1.22\ \mu$s

   (c) Where does the charge go?

   The conduction current carries it to the surface of the material, where it stays. The total charge does not change.

Documentation: _____